

Topic – 1 (Introduction to Python)

1. Write a python program to print a message on screen.
2. Write a python program to take two numbers as input now find summation, subtraction, multiplication and division of two numbers.
3. Write a python program to find area and perimeter of circle.
 $area = 3.14 * r * r$
 $perimeter = 2 * 3.14 * r$
4. Write a python program to find volume and surface area of cuboid.
 $v = l * b * h$
 $sa = 2 * (l * b + b * h + h * l)$
5. Write a python program to find simple interest.
 $si = (p * n * r) / 100$
6. Write a python program to find square root of a number.
 $root = \text{math.sqrt}(n)$
7. Write a python program to take number of days as input now display years, weeks and days. E.g. If user input 375 days then output will be 1 years, 1 weeks and 3 days. In this program ignore leap year.
8. Write a python program to compute area of perimeter of rectangle.
 $area = l * b$
 $perimeter = 2 * (l + b)$

Topic – 2 (Flow control in Python)

1. Write a python program to check given number is even or odd.
2. Write a python program to find greatest number in two numbers.
3. Write a python program to find roots of quadratic equation $ax^2 + bx + c = 0$.
 $root1 = (-b + \text{math.sqrt}(b^2 - 4 * a * c)) / (2 * a)$
 $root2 = (-b - \text{math.sqrt}(b^2 - 4 * a * c)) / (2 * a)$
4. Write a python program to take coordinates of a point as input and determine its quadrant.
5. Write a python program to compute gross salary based on following parameters:-

Basic Salary	HRA	DA
1-4000	10%	50%
4001-8000	15%	60%
8001-12000	20%	70%
More than 12000	25%	80%

$gs = bs + hra + da$

6. Write a python program to calculate electricity bill based on following parameters:-

Units	Bill/unit
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1-150	2.40
For next 151-300	3.00
For next more than 300	3.20

7. Write a python program to print table of given number in given format:-

2*1=2

2*2=4

2*3=6

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2*10=20

8. Write a python program to find factorial of given number.
9. Write a python program to find sum of digits of given number.
10. Write a python program to reverse the digits of given number.
11. Write a python program to convert binary number to its decimal equivalent.
12. Write a python program to check given number is prime or not.
13. Write a python program to print series of prime numbers in given range. Range must be created by taking input from user.
14. Write a python program to print Fibonacci series up to n terms, where value of n is entered by user.
15. Write a python program to check given number is Armstrong or not.